Retail Sales Forecasting Using Machine Learning

Student: [Your Name] | Semester: Fall 2026 | Course: CPSC-59000

Project Topic and Motivation

This project will investigate how accurately future retail sales can be predicted using historical sales data. Retailers need reliable forecasts to plan inventory, staffing, and promotions. However, sales often change with holidays, seasonal demand, and economic conditions. I want to explore these patterns and compare simple forecasting methods with machine learning models to understand which approaches work best for a real-world retail dataset.

Dataset and Preliminary Research

I plan to use the publicly available Walmart Store Sales Forecasting dataset on Kaggle, which includes historical weekly sales and related information such as holidays and economic indicators. Prior work on forecasting provides a foundation for the project. Hyndman and Athanasopoulos [1] explain time-series forecasting methods and evaluation. The M5 forecasting competition [2] demonstrates the challenges of forecasting retail demand at scale. The Walmart dataset description [3] provides the data context for the proposed analysis.

Proposed Methodology

First, I will inspect and clean the dataset, handle missing values, and explore sales trends by store, time period, and holiday. Next, I will create time-based features and use a chronological training/test split to avoid using future information in model training. I will establish a seasonal-naive baseline, then compare it with a statistical time-series model and a machine learning approach such as Random Forest or gradient boosting. Forecast accuracy will be evaluated using MAE and RMSE on held-out weeks. Finally, I will visualize actual versus predicted sales, analyze major forecasting errors, and summarize practical findings and limitations.

Expected Outcome

The final deliverables will include a reproducible Python notebook, visualizations of sales patterns and forecast performance, and a written report comparing models. The goal is to identify a practical forecasting approach rather than assume that a complex model will always perform better.

References

[1] R. J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed., OTexts, 2021. https://otexts.com/fpp3/

[2] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, “M5 accuracy competition: Results, findings, and conclusions,” International Journal of Forecasting, vol. 38, no. 4, pp. 1346–1364, 2022.

[3] Kaggle, “Walmart Recruiting – Store Sales Forecasting,” dataset and competition description. https://www.kaggle.com/competitions/walmart-recruiting-store-sales-forecasting